WEEK 4 · DAY 1 · INSTRUCTOR NOTES

Scoring a Detector + Quiz 2

For the trainer only.

The one idea for today: *a detector is only as good as its score against the answer key, and the score must be recall and precision, with the rules written down first.* Accuracy is banned for rare events.

At a glance

Length	~2 hours, including Quiz 2 (20 min + 10 min marking)
Participants leave able to	score any detector at event level, explain recall vs precision, refuse to quote accuracy, turn false alarms into engineer hours, say why rain fade isn't an anomaly
Covers (original)	Week 4 Code Book: *5. Anomaly ground truth*, *Scoring conventions*, *A reusable event-level scorer*; Week 4 quiz topic *evaluating a detector* (all 6 questions); Quiz 2 = the whole Week 4 MCQ file
Datasets	`week-04/data/telemetry_anomaly_labels.csv` (repaired on load), `bus_power_telemetry_hourly`, `twta_thermal_daily` (week-04), `battery_health_daily` (week-04), `rf_link_quality_hourly`
Files you need open	animated slides, your **solution** notebook, `W4D1-quiz-2-answer-key.pdf`

The day before: checklist

☐ Run the **solution** notebook top to bottom: 14 of 14 ✅ (about 30 seconds).

☐ Print `2-exercise/quiz/W4D1-quiz-2.pdf` (one per person), **or** import `W4D1-quiz-2-google-forms.csv` into Google Forms / your LMS.

☐ Print or open `4-instructor-only/W4D1-quiz-2-answer-key.pdf`. Don't share it until marking.

☐ Remind the room: today the answer key is revealed, so anyone still finishing W3 detectors should do it before class.

Run of show (2 hours)

Time	Block	What happens	Slides
0:00	**Hook**	Musa vs Ade the mai-guards; vote; the 99.87% accuracy trap	1–4
0:10	**The answer key**	Six events on three lanes; the four scoring rules; recall vs precision	5–8
0:25	**Report cards**	Each detector scored on screen; the full report card	9–15

Time	Block	What happens	Slides
0:35	**Count the cost**	False alarms → hours; rain fade isn't an anomaly; recall-first vs precision-first	16–19
0:40	**Hands-on**	Parts A, B, C (50 min). Walk the room.	21
1:30	**Quiz 2**	20 minutes, phones down, no AI	20
1:50	**Mark + wrap**	Swap and mark (letters from the key), re-teach any question a third of the room missed, recap	22

If time is tight, run the quiz at the start of W4D2 instead.

Explain it like this (in plain English)

- **Why an answer key?** Without it, "my detector works" is an opinion. With it, it's a number.
- **Event-level scoring:** we care whether we *noticed* each problem, not how many rows we painted red. One flag inside the event = caught; many flags = still one catch.
- **False alarm:** every flag outside a real event. Think of each one as a phone call to the duty engineer at 2 a.m.
- **Recall:** "Did we miss anything?" **Precision:** "When it pages me, is it real?" You always need both.
- **Accuracy lies** for rare events: 6 events in 4,706 readings means "always say no" scores 99.87%.
- **Conditioning beats widening:** the all-hours chart pages on every eclipse hour (normal physics). Add "only judge sunlit hours" and the false alarms vanish without losing a single catch.
- **Hours, not ratios:** "precision 0.06" means nothing to a manager. "31 hours a year of pointless pages, about 4 shifts" does.
- **Unusual ≠ anomaly:** rain fade is rare and dramatic, but it's physics with a known cause. Alarm on the *service* (margin below 0 dB) and use a rain gauge to explain it.

The report card (computed from the data, same scorer for all)

Detector	Watches	Caught	False alarms	Recall	Precision
State-of-charge floor < 82%	bus power (EV01–03)	0/3	18	0.00	0.00
3σ control chart, all hours	bus power	3/3	46	1.00	0.06
3σ control chart, sunlit only	bus power	3/3	0	1.00	1.00
Raw 3σ on temperature	TWTA (EV04–05)	0/2	0	0.00	– (never fired)
Residual 3σ (drive + season + age, fitted on 2024)	TWTA	2/2	0	1.00	1.00
CUSUM on the residual (k = 0.5, h = 5)	TWTA	2/2	2 (21 Apr, 6 May 2025)	1.00	0.50

Detector	Watches	Caught	False alarms	Recall	Precision
Fixed floor 95 Ah	battery (EV06)	0/1	0	0.00	– (never fired)
🌶️ 90-day fade rate below −3 Ah/yr	battery	1/1	0	1.00	1.00 (first alarm 2 May 2025, 31 days after the knee)

Supporting numbers: sunlit array current mean 28.21 A, sd 0.594 A, band 26.43–29.99 A; all 46 false alarms are at 21:00 UTC (eclipse hour); the 3 dropouts read about 15.3–15.5 A. TWTA raw 3σ ceiling 69.8 °C vs highest day 68.2 °C (the EV04 spike); residual sd 0.378 °C; EV04 is 13σ in residual terms. Battery lowest day 95.39 Ah (never below 95).

Slide talking points

#	Say this	Ask this
3	"Neither guard is good. Today we learn the two numbers that prove it."	"Who would you hire?" (most pick Musa)
4	Point at the 6 red dots among 4,706.	"A detector that does nothing scores how much?"
7	Read the 4 rules aloud; get a thumbs-up from the room *before* anyone computes.	"Does flagging one day of EV05 count?"
10	Every red dot is a normal eclipse night: the battery doing its job.	"Is this alarm wrong or useless?" (useless)
11	Grey dots = eclipse, gold dots = real dropouts. One condition removes all the grey.	–
12	The raw series swings 7 °C a year, so a 2 °C shift hides inside it. The residual removes the swing.	–
13	CUSUM is more sensitive. Here that bought 2 false alarms and nothing else.	"So is CUSUM bad?" (no: it wins when the shift is small vs the noise)
14	No single battery day is strange; the *speed* changed.	"What kind of alarm would catch that?"
17	46 × 2 × 20 min ≈ 31 h.	"What's your unit's real minutes-per-alarm?"
18	Only 12 of the 18 flagged hours were real outages; all 18 were rainy.	Quiz Q6: True or False?
19	NigComSat-1 (2008, solar-array failure): why power systems are recall-first.	–

Answer key

Task	Answer	Common mistake → what to say
A1	`1 - n_events / n_readings` = 0.9987	`n_events / n_readings` (that's the error rate)
A2	`"yes"` , `1`	Answering 2 for EV02 (rule 2: many flags = one catch)
A3	`9 / 10` = 0.90, `9 / 200` = 0.045	Swapping them: recall is about *real events*, precision about *alarms*
B1	`SOC_FLOOR = 82` → caught 0/3, 18 false alarms	Expecting it to catch the dropouts: it's watching the wrong column
B2	`K = 3` → 3/3, 46 false alarms	–
B3	`== 0` → 3/3, 0 false alarms	Writing `== 1` (that keeps only eclipse hours)
B4	`N_SIGMA = 3` → 2/2, 0 false alarms	–
B5	`H = 5` → 2/2, 2 false alarms	Other H values give different counts; the check expects the standard 5
B6	`"EV06"`	–
B7	`3`	–
C1	`false_alarms_per_season * seasons_per_year * MINUTES_PER_FALSE_ALARM / 60` = 30.7	Forgetting `/ 60` (gives minutes)
C2	`n_real / n_flags` = 12/18 = 0.67	–
C3	`False`	Writing `"False"` with quotes is also accepted
🌶️	`RATE_LINE = -3.0` (anything that catches EV06 with 0 false alarms: below about −2.6 and above about −3.9)	Picking −2: fires on normal fade too

Quiz 2

- **Files:** `2-exercise/quiz/W4D1-quiz-2.pdf` (printable, no answers), `W4D1-quiz-2-google-forms.csv` (import), `4-instructor-only/W4D1-quiz-2-answer-key.pdf` .
- **Source:** the original `Time-series/MCQ/week-04-mcq.csv` (that file is unchanged), with three questions edited: Q2 and Q23 corrected to match the data, Q4 renumbered to our Week 3. The answer key lists them. Options reshuffled with a fixed seed (4104) so the correct letters are spread: multiple choice A ×7, B ×7, C ×6, D ×6; True/False A ×2, B ×2. (In the original file, most answers were option B.)
- **Topic → day:** alarm software → W3D1 · SPC and residuals → W3D2 · change points and drift → W3D3 · evaluating a detector → today.
- **Two questions are corrected to match the data** (their answers are unchanged):

- **Q2**: the original said the SoC < 82% floor "fires on all 46 eclipse hours". It fires **18** times, all on eclipse hours, and the question now says so. The detector that fires on all 46 eclipse hours is the all-hours control chart (slide 11).
- **Q23**: the original said the fade rate "roughly doubled". The question now gives the real numbers: about **−2.2 → −3.7 Ah/year** (×1.6).

Troubleshooting

Problem	Fix
`ParserError: Expected 8 fields in line 5, saw 9`	Someone loaded the key with `pd.read_csv`. Use `load_labels()` (STEP 0)
`ValueError: Invalid frequency: H` / bad tolerance	Old code using `"1H"`. Use `"1h"`
Scores of `events: 0` / `ZeroDivisionError`	The `source_file` string doesn't match the key exactly; copy it including `.csv`
A participant's B5 gives different false alarms	They changed k or H. The check expects k = 0.5, H = 5
Fast finishers	🌶️ slope alarm, then: "try K = 2 in B2 and B3: what happens to precision?" (sunlit-only jumps to 138 false alarms at K = 2)

Notes on the original material

- The original labels file **crashes on load** (`ParserError`): EV04's cause text has an unquoted comma. Our `load_labels()` / `common.read_labels()` repair it; the original file is untouched.
- The original scorer uses `tol="1H"`, which pandas 3 rejects. Ours uses `"1h"` / `"1D"`.
- The original code book (section 4) says a naive 3σ rule on C/N flags "roughly all 56 rainy hours, a 79% false-alarm rate". The data says **18** flags, **12** real, **6** false (33%). Slide 18 and C2 use the real numbers.
- The original code book (section 1) says the sunlit standard deviation is about 0.06 A. The data says **0.594 A** (band 26.43–29.99 A). The dropouts (~15.5 A) are still far outside it.
- The original's worked example scores a state-of-charge floor against *solar-array current* events: we keep that on purpose, because it shows a detector watching the wrong channel.